INCH-POUND
MIL-DTL-24643/64B
27 November 2023
SUPERSEDING
MIL-DTL-24643/64A
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/64A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TWISTED SHIELDED PAIR(S), -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSTPSJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight)

First	-	Copper conductor, tin-coated, in accordance with ASTM B286 (see table I for sizes).
Second	-	Separator may be used at manufacturer's option where required to provide free stripping insulation.
Third	-	Thermoset insulation. Nominal 0.010 inch (see table I for finished wire diameter).
Fourth	-	Two conductors (one black and one white) twisted together with a lay length as specified in table II. Twisted pair identification code applied by method 6 if more than one pair in cable.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 36 AWG tin-coated copper. Minimum coverage shall be 90 percent and the braid angle shall be between 20 and 40 degrees.
Seventh	-	The required number of pairs (see table III) cabled together with a lay length as specified in 3.4.4 of MIL-DTL-24643. Fillers may be used as required to form a firm, well-rounded assembly. Cabling sequence shall be consecutive for four pair cables.

NOTE: For one pair cable, omit the seventh step.

Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. Jacket wall thickness shall be 0.030±0.005 inch for cables with core diameters less than 0.300 inch. Jacket wall thickness shall be 0.040±0.005 inch for cables with core diameters greater than or equal to 0.300 inch (see table III for finished cable diameter).

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

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TABLE I. Wire configurations and dimensions.

Conductor size (AWG)	Stranding (inches)	Conductor diameter (inch) (±0.003 inch)	Finished wire diameter (inch)		Conductor resistance (ohms)
			min.	max.	
24	19 x 36	0.024	0.041	0.047	25.91
22	19 x 34	0.030	0.047	0.053	16.21
20	19 x 32	0.039	0.056	0.062	9.95
18	19 x 30	0.049	0.066	0.072	6.34
16	19 x 29	0.056	0.073	0.079	4.92
14	19 x 27	0.070	0.087	0.093	3.11

TABLE II. Length of lay.

Wire size range	Lay length	Twist tolerance (inch)
20-24	1	±0.25
14-18	1.5	±0.25

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TABLE III. Finished cable details.

Military part no. M24643/64	Type and size	Conductor size (AWG)	Number of pairs	Finished cable diameter (inch) (±0.016 inch for single pair cables) (±0.030 inch for multi-pair cables)
-01UN	LS1TPSJ-24	24	1	0.170
-02UN	LS1TPSJ-22	22	1	0.182
-03UN	LS1TPSJ-20	20	1	0.200
-04UN	LS1TPSJ-18	18	1	0.220
-05UN	LS1TPSJ-16	16	1	0.234
-06UN	LS1TPSJ-14	14	1	0.262
-07UN	LS2TPSJ-24	24	2	0.260
-08UN	LS2TPSJ-22	22	2	0.285
-09UN	LS2TPSJ-20	20	2	0.325
-10UN	LS2TPSJ-18	18	2	0.390
-11UN	LS2TPSJ-16	16	2	0.420
-12UN	LS2TPSJ-14	14	2	0.465
-13UN	LS3TPSJ-24	24	3	0.275
-14UN	LS3TPSJ-22	22	3	0.285
-15UN	LS3TPSJ-20	20	3	0.325
-16UN	LS3TPSJ-18	18	3	0.390
-17UN	LS3TPSJ-16	16	3	0.450
-18UN	LS3TPSJ-14	14	3	0.490
-19UN	LS4TPSJ-24	24	4	0.295
-20UN	LS4TPSJ-22	22	4	0.325
-21UN	LS4TPSJ-20	20	4	0.395
-22UN	LS4TPSJ-18	18	4	0.450
-23UN	LS4TPSJ-16	16	4	0.480
-24UN	LS4TPSJ-14	14	4	0.540

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to shield	1,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75

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Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation) (cycles, min.) (applicable to multi-pair cables only)	25
Permanence of printing (jacket) (cycles, min.)	125
Shield (conformance to material, constructions, and coverage)	No failure
Surface transfer impedance (type LS1TPSJ only)	
Milliohms per meter (max.)	Not applicable
Electromagnetic pulse (EMP) response (dB, min.)	Not applicable

Group D:

Flame propagation (finished cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure
Durometer (jacket) – type A (hardness, min.)	80

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTPSJ (all sizes)	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-080)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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